

Experiment 26: pH of materials around the home

Notes

Our environment contains a variety of solutions. The most obvious being oceans, lakes and rivers but also including soil moisture and ground water. Some of these may become contaminated by the deliberate or inadvertent addition of substances that are in common usage in industry, agriculture or in the home. One of the effects of this contamination is to change the pH of these solutions.

The pH of solutions can be measured using indicators and electronic pH meters. Indicators are materials that change colour in different pH solutions. Universal indicator is a commonly used indicator. It is a mixture of several indicators each exhibiting different colours in acids and bases. Its colour results from the combination of the colours of all the indicators present. Because it exhibits a range of colours over a large range of pH values, universal indicator can be used to estimate the pH of solutions.

In this experiment you will test the pH, using universal indicator, of a variety of materials found around the home and used and released into the environment.

Equipment

test tubes (several)

universal indicator solution (5 mL)

universal indicator colour chart

samples of:

washing powder

household ammonia

baking soda

sugar

salt

orange juice

lemon juice

lemonade

vinegar

milk

aspirin

antacid preparation

soluble garden fertiliser

garden or builder's lime

drain or oven cleaner

tooth paste

mouth wash



Procedure

SAFETY NOTE:

- Garden or builder's lime and drain or oven cleaner should not be allowed to come into contact with your skin.
- Handle these substances with extreme care.

1. Preparation of samples.
 - (i) Clear solutions can be tested directly.
 - (ii) Solutions that have solids mixed with them should be decanted or filtered.
 - (iii) Solids should be mixed with water, stirred thoroughly and then the solution should be treated as in (i) or (ii) as appropriate.
2. Place about 2 mL of each solution into separate test tubes.
3. Add two drops of Universal Indicator TM and, using the colour chart provided with the indicator, note the pH of the solutions and record your results in a suitable table.

Processing of results and questions

1. List all those solutions that were neutral.
2.
 - (a) List all those solutions that were acidic.
 - (b) What is the name of the compound in each of these that is responsible for the solutions being acidic? You may have to research some of these.
 - (c) Choose one of these substances and write a balanced ionic equation to show why the solution is acidic.
3.
 - (a) List all those solutions that were basic.
 - (b) What is the name of the compound in each of these that is responsible for the solutions being basic? You may have to research some of these.
 - (c) Choose one of these substances and write a balanced ionic equation to show why the solution is basic.
4. Investigate the pH of rainwater and groundwater in your area.
 - Predict the pH you expect the water to have.
 - Explain your predictions.
 - Measure and record the pH readings of the rainwater and groundwater.
 - Explain any differences between predicted pH and measured pH values.

Notes